



DPUTablePlayer

DPUTablePlayer plays values from a table that has been stored in CSV format. In contrast to DPUSampleFilePlayer, the values are not played sequentially during the simulation. Instead, it can be defined for each column by an input, which value is currently played.

1. Inputs:

<i>Start</i>	If the value is set to 1 the DPU starts playing values from file.
<i>Stop</i>	If the value is set to 1 the DPU stops playing values from file. The current position is saved. The next time <i>Start</i> is set to 1.0 the current position is used.
<i>Reset</i>	If the value is set to 1 the DPU resets the position within the data file to the first value.
<i>[Name]Row</i>	For every column of the table with the name <i>[Name]</i> an input <i>[Name]Row</i> exists. This input determines from which table row the value, which is output in output <i>[Name]</i> , is taken. The first table row, which contains values, has the index 0. If <i>[Name]Row</i> is smaller than 0 bigger than the index of the last table row, the default value is output.
<i>Offset</i>	The <i>Offset</i> value is added to all line numbers, which are defined by the variables <i>[Name]Row</i> , to determine the table row from which the value is taken.

2. Static parameters:

<i>TableFile</i>	Filename of table file. If no absolute path is specified, the file is searched below the SILAB installation directory.
<i>ColumnSep</i>	The character, which is used in the CSV file as column separator. Default: ";"
<i>DecimalSep</i>	The character, which is used in the CSV file as decimal separator. Default: ","
<i>UseDefault</i>	If the parameter <i>UseDefault</i> is set to <i>true</i> , table elements may be left blank and a default value will be returned. Otherwise, while reading the table, an error is issued when an empty field is discovered. Default: <i>false</i> .
<i>Default</i>	If a table element is left blank, this default value is returned. Default: 0
<i>Wrap</i>	If the parameter <i>Wrap</i> is set to <i>true</i> , the content of the table is repeated infinitely. If a negative or too big line is to be returned, then



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the line number modulo number of lines is used instead. Therefore, the default values are no longer used. Default: *false*.

3. Outputs:

[Name]

For every column of the table with [Name] an output with the same name exists. This output variable contains the content of the selected table element.

MaxRows

Number of rows of the table file.

4. Functional principle:

First, create a table in a spreadsheet program, e.g.:

	A	B	C
1	Dist	Speed	
2	1	0,2	
3	5	3,1	
4	7	4,4	
5	4	-9,9	
6	11	-1	
7	23	-3	
8	45	-5,5	
9	69	3,3	
10			
11			

Save the table as CSV file, e.g. under the name „D:\SILAB\Data\TablePlayer.csv“. In the German Version of Microsoft Excel the file looks as follows:

```
Dist;Speed
1;0,2
5;3,1
7;4,4
4;-9,9
11;-1
23;-3
45;-5,5
69;3,3
```

Column separator

Decimal separator

The semicolon is used as column separator, and the comma is used as decimal separator.



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Now include the DPUPTablePlayer as follows:

```
DPUPTablePlayer player
{
    Computer = {MAIN};
    Index = 1;

    Default = -9999;
    UseDefault = true;

    TableFile = "Data\TablePlayer.csv";
    ColumnSep = ";";
    DecimalSep = ",";
};
```

Connect inputs and outputs:

```
Connections =
{
    ... <-> player.DistRow,
    ... <-> player.SpeedRow,
    player.Dist <-> ...,
    player.Speed <-> ...
};
```

Now, if variable *DistRow* has the value 2 output *Dist* has the value 7 (= Content of line number 2 (counting from 0) = 3rd line of column *Dist* in the table).